

R-2 Capsule Launcher

Assembly and maintenance instructions

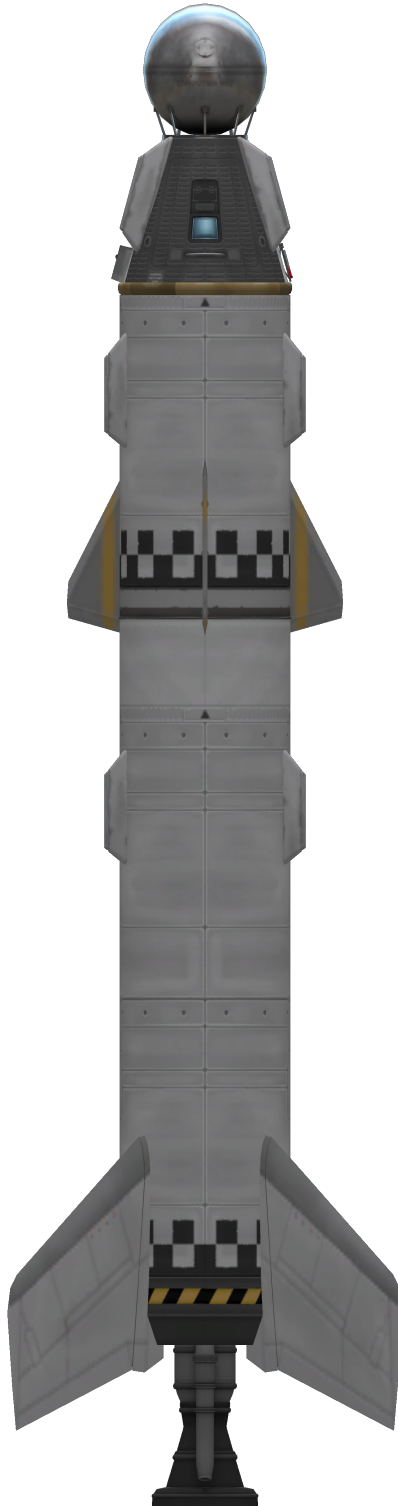


Table of Contents

Description..... 3

Basic construction..... 3

Assembly instructions..... 3

 Parts list..... 3

 Technology..... 3

 Stacking..... 3

 Stage configuration..... 6

 Lower stage..... 6

 Upper stage..... 6

 Staging..... 6

Post flight maintenance..... 7

Pre rollout checks..... 7

Description

The R-2 is a dumb, multi-stage, recoverable, liquid fuelled booster for launching naked capsules based on the Mk-1 pod.

The rocket is composed of pre-assembled stages and is relatively simple to construct once these assemblies are ready.

No computer is provided. The R-2 takes its commands from the payload. The lower stage provides control via movable fins. The upper stage engine is gimballed.

The whole system is recoverable, with the exception of the upper stage fins which must be replaced after every orbital flight.

Basic construction

The rocker is built in a single assembly operation and construction is straightforward. There is only a single model of the R-2 rocket, comprised of the RS-1A booster and the SM-1 upper stage.

The launcher is designed for launching AOC-0 and AOC-1 capsules. The R-2 provides launching with enough delta V for deorbiting and minimal to no maneuvering (from at least 100 m/s for a polar orbit to at least 300 m/s for an equatorial one).

Assembly instructions

Parts list

- 1x RS-1 lower stage
- 1x SM-1 upper stage

Technology

The rocket requires a level 4 technology level. Both stages require this technology level.

Stacking

Stack SM-1 over RS-1. Connect SM-1 engine to RS-1 decoupler. Capsule should be attached to upper stage decoupler.

When stacking, the following rules apply.

- Shroud **MUST** be enabled
- Parachutes **MUST** all have the same orientation (N/S by default)
- RS-1 gas generator **MUST** should point in the same direction as one of the upper stage fins
 - This is the “up” direction
- Lower stage fins **MUST** be oriented forming an X when compared with up/down and side directions

- They MUST NOT point in the up/down/left/right directions

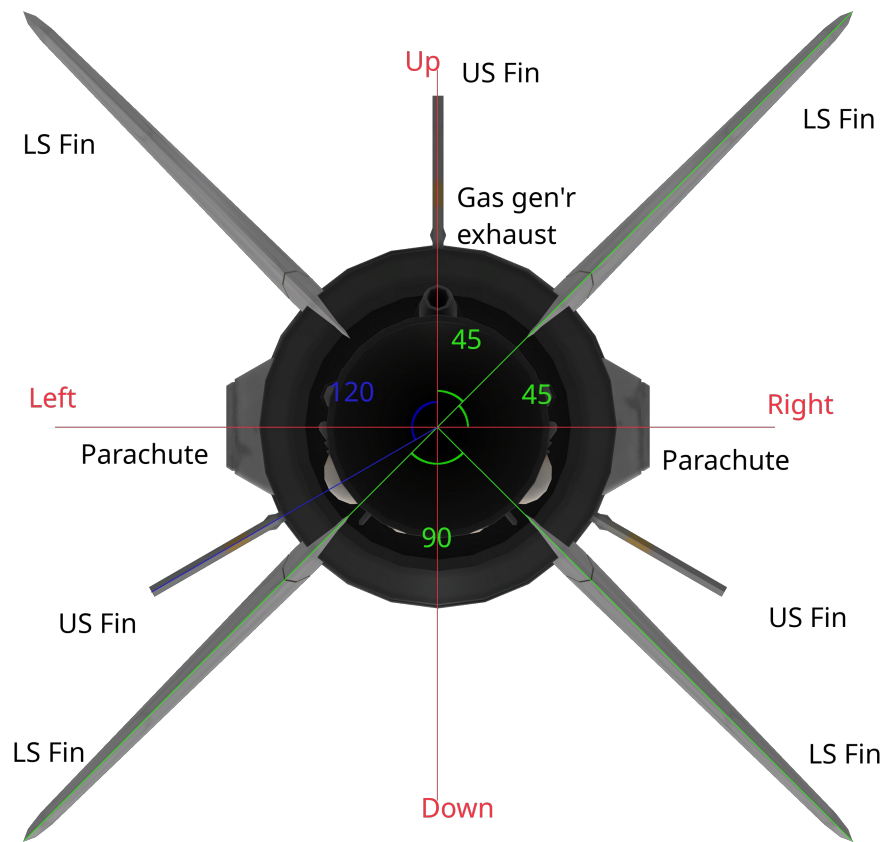


Figure 1: Relative positions of parts in stacked launch vehicle as viewed from the nozzle



Figure 2: R-2 800 final stacked form side view

Stage configuration

Configuration is derived from stage configuration. Tables below are copied from their respective manuals.

Lower stage

Part	Variable	Value	Default	Notes
Fins	Pitch	Active	Yes	
	Yaw	Active	Yes	
	Roll	Active	Yes	
	Deploy	Retracted	Yes	
	Deploy direction	Normal	Yes	
	Deploy angle	0	No	
	Authority limiter	4	No	
	Throttle	Main throttle	Yes	
Engine	Thrust limiter	100	Yes	Uppermost tank
	Flow priority	0	Yes	
LFOX tank 1	Flow priority	+1	No	
LFOX tank 2	Flow priority	+2	No	
LFOX tank 3	Flow priority	+3	No	If present This tank is not used in the R-2
LFOX tank 4	Flow priority			
Decoupler	Force percent	100	Yes	
	Enable crossfeed	False	Yes	
	Enable staging	True	Yes	
Parachutes	Deploy mode	When safe	Yes	
	Min pressure	0.45	No	
	Altitude	500	No	
	Spread angle	10	No	

Upper stage

Part	Variable	Value	Default	Notes
Engine	Thrust limiter	100	Yes	
	Throttle	Main throttle	Yes	
LFOX tank	Flow Priority	0	Yes	
Decoupler	Force percent	100	Yes	
	Enable crossfeed	False	Yes	
	Enable staging	True	Yes	
Parachutes	Deploy mode	When safe	Yes	
	Min pressure	0.45	No	
	Altitude	500	No	
	Spread angle	10	No	

Staging

Stage	Parts	Notes
0	US Parachutes, US Decoupler	
1	US Engine, LS Parachutes, LS decoupler	
2	LS Engine	

Post flight maintenance

When stacking after a flight, the following abbreviated maintenance procedure can be executed. If in doubt, execute full stacking procedure.

- RS-1 Decoupler
 - Enable staging
- SM-1 Decoupler
 - Enable staging
- Tanks
 - Fill

Pre rollout checks

- Verify staging
- Verify parachute settings
- Verify decoupler staging
- Verify payload is present